R13

Code No: 117DX

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year I Semester Examinations, July/August - 2023
INFORMATION RETRIEVAL SYSTEMS
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

PART – A

(25 Marks)

1.a)	What is a Language Model?	[2]
b)	Write a brief note on Retrieval Strategies.	[3]
c)	What are Retrieval Utilities?	[2]
d)	What is the use of Relevance Feedback?	[3]
e)	What is R-Distance?	[2]
f)	Define CLIR.	[3]
g)	What are Shingles?	[2]
h)	Write a brief note on Vector Space Simplifications in Query Processing.	[3]
i)	List the CURD operations of RDBMS.	[2]
j)	What is the need of portability of DBMS?	[3]

PART – B

(50 Marks)

2. How to measure similarity between query and document in Vector Space Model? Explain with example. [10]

OR

3. Describe how the individual term estimates can be combined into a total estimate of relevance for the document. [10]

4. Why relevance feedback has become popular information retrieval utility? Explain its process in detail. [10]

OR

5.a) Explain Regression Analysis.
b) How to use thesaurus to expand a query? [5+5]

6.a) Describe parts-of-speech tagging and word sense tagging for identifying phrases.
b) Contrast information retrieval with information extraction. [5+5]

OR

7.a) Explain Resources in Cross-Language Retrieval Systems.
b) Compare structured queries and balanced queries. [5+5]

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8. Explain building an Inverted Index and compressing an inverted index. [10]

OR

9. Demonstrate construction of a signature to a document and finding false positives.[10]

10. Explain Information Retrieval as an application of RDBMS. [10]

OR

11.a) How to compute relevance using unchanged SQL?

b) Describe Page Rank algorithm for improving Web search. [5+5]

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